

Analysis of Variance

CSU, Chico Math 351

2026-04-29

outline

Recap

ANOVA - The Basics

- definition

- intuition

- assumptions

ANOVA - The Details

- test statistic

- F-distribution

Example

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recap, Two Sample t -test

The two sample t -test compares the means of two groups in the hypothesis:

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2.$$

This test required a new statistic with a new calculation for standard error.

recap, Two Sample t -test

The two sample t -test formulas

$$t_{df} = \frac{(\bar{X}_a - \bar{X}_b) - (\mu_a - \mu_b)}{s_{\bar{X}_a - \bar{X}_b}} \quad \text{and} \quad (\bar{X}_a - \bar{X}_b) \pm t_{df}^* s_{\bar{X}_a - \bar{X}_b}$$

where

$$s_{\bar{X}_a - \bar{X}_b} = \sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}$$

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Analysis of Variance, motivation

If there were three or more groups, the two sample t -test would not work. We could force the test on the data by comparing two groups at a time, but this has dangerous implications¹. We thus require a new statistical method, **analysis of variance**, or **ANOVA**.

¹We'll go into more detail on this when discussing the Bonferroni and Tukey methods for multiple comparisons.

Analysis of Variance, definition

ANOVA

Analysis of variance uses a single hypothesis to check whether the *means* across two or more groups are equal, and uses the F statistic to evaluate this hypothesis.

ANOVA, hypotheses

The ANOVA hypotheses for k groups are

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

H_A : at least one mean is different.

ANOVA, hypothesis examples

With ANOVA you can compare the means by groups for many different data sets.

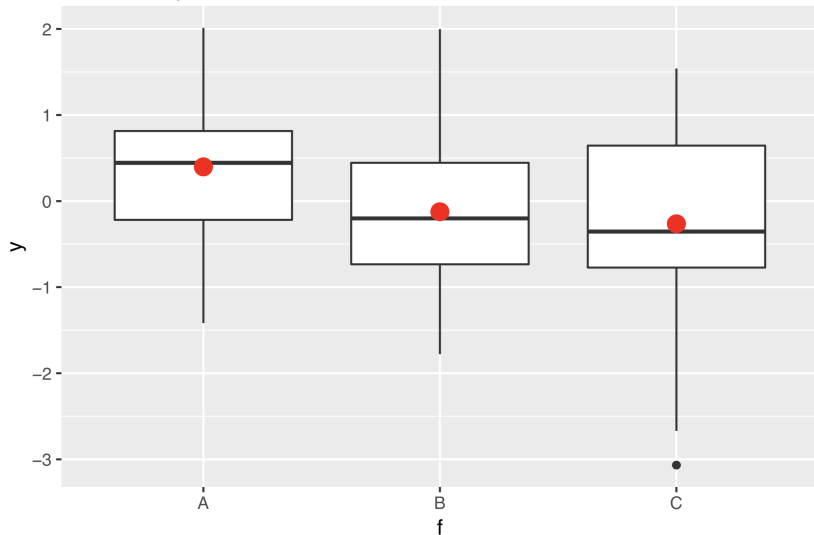
- ▶ mean batting average by position,
- ▶ mean movie budget by genre, (or year),
- ▶ mean CO₂ intake by treatment, or location, (or concentration)
- ▶ mean REM sleep by conservation status,
- ▶ mean birth/body weight by family,
- ▶ ... numerical variable by levels of a categorical variable ...
- ▶ ...

ANOVA, intuition

Analysis of variance tells us about means by groups, despite its name. Large variation **among** the groups relative to small variation **within** the groups indicates different population means.

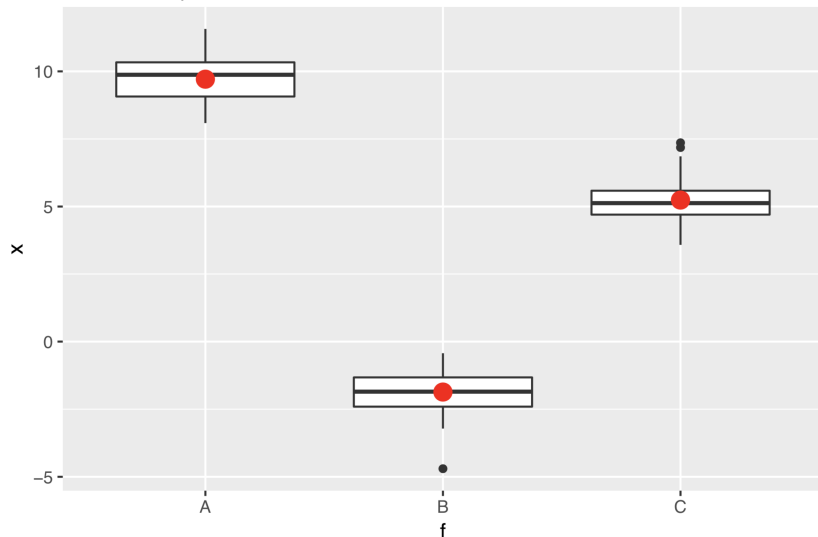
ANOVA, intuition via picture

What do you think of these means (think of variation within and among groups)?



ANOVA, intuition via picture

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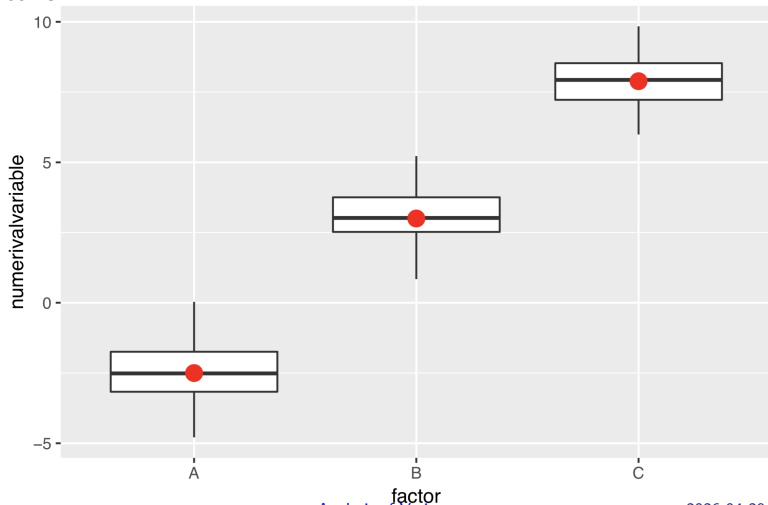
ANOVA, assumptions

The ANOVA has three basic assumptions,

- ▶ independent observations (among and within groups)
- ▶ data within each group are nearly normal, and
- ▶ the variability of each group is about equal.

ANOVA, picture

The last two assumptions can help us visualize just what is going on. It starts with box plots by groups—what other indicators may reveal whether the groups in question have similar or different means?



ANOVA, test statistic

ANOVA calculates one fraction based on two numbers, variation among (between) groups and variation within groups. These two numbers are generally referred to as mean square values.

mean squared variation among groups

MSG is a strictly positive measure of the variation across all groups, and has $df_G = k - 1$ where k represents the number of groups.

mean squared error

MSE is a strictly positive measure of the variation within groups, and has $df_E = n - k$ where k represents the number of groups and n is the sample size.

We'll see the calculations for these values in the following section.

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Premise

Back when we covered two-sample hypothesis tests, we saw the F-test used for comparing variances. The test statistic there was the ratio of sample variances: $F = \frac{\sigma_2^2}{\sigma_1^2}$. This approach forms the basis of Analysis of Variance (ANOVA).

If we wish to compare the average effects for k different levels of a categorical variable, we will effectively have k independent random samples of sizes $n_1, n_2, \dots, n_k$, with total sample size: $n = \sum_{j=1}^n n_j$.

Dot Notation for Totals/Mean

Additionally, we let Y_{ij} represent the i th entry for the j th level. To indicate the total for the j th sample, we use dot notation—including a dot to represent summing over a given subscript:

$$T_{.j} = \sum_{i=1}^{n_j} Y_{ij} = Y_{1j} + Y_{2j} + \dots + Y_{n_j j}$$

for which the sample mean would be:

$$\bar{Y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} \frac{T_{.j}}{n_j}$$

By extension, the OVERALL total and mean will be:

$$T_{..} = \sum_{i=1}^k \sum_{j=1}^{n_j} Y_{ij} = \sum_{i=1}^k T_{.j} \quad \bar{Y}_{..} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_j} Y_{ij} = \frac{1}{n} \sum_{i=1}^k T_{.j}$$

Assumptions

To test whether all levels' means are equal, we make an assumption: the Y_{ij} s are independent and normally distributed with mean μ_j ($j = 1, \dots, k$) and variance σ^2 (constant for all j). The resulting model equation is:

$$Y_{ij} = \mu_j + \epsilon_{ij}$$

with ϵ_{ij} representing the difference between observed Y_{ij} and its expected value, where $\epsilon_{ij} \sim N(0, \sigma^2)$.

We can also let the overall average across all n observations be μ , where $\mu = \frac{1}{n} \sum_{j=1}^k n_j \mu_j$.

SSG - Sum of Squares among Groups

Since $\bar{Y}_{..}$ is a natural estimate for μ ,

$$SSG = \sum_{j=1}^k \sum_{i=1}^{n_j} (\bar{Y}_{.j} - \bar{Y}_{..})^2 = \sum_{j=1}^k n_j (\bar{Y}_{.j} - \bar{Y}_{..})^2$$

This value considers the sum of squared differences between each group's sample mean and the overall mean. As the differences among groups increase, so does SSG.

Additionally, when μ_j for all k groups are all the same, $E(SSG) = \frac{(k-1)}{\sigma^2}$. Therefore, $\frac{SSG}{\sigma^2}$ follows a Chi-Squared distribution with $k-1$ degrees of freedom. However, σ^2 is frequently unknown and needs to be estimated.

SSE - Sum of Squares Error (within Groups)

The j th sample variance is written:

$$S_j^2 = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} (Y_{ij} - \bar{Y}_{.j})^2$$

By multiplying each sample variance by $n_j - 1$ and summing over j , we get the numerator of the pooled estimator for σ^2 , known as SSE:

$$SSE = \sum_{j=1}^k (n_j - 1) S_j^2 = \sum_{j=1}^k \sum_{i=1}^{n_j} (Y_{ij} - \bar{Y}_{.j})^2$$

$\frac{SSE}{\sigma^2}$ has a Chi-Square distribution with $n - k$ degrees of freedom, and SSE and SSG are independent. From here, the TOTAL sum of squares (SST) is the sum of SSG and SSE.

The F-test

If $H_0 : \mu_1 = \dots = \mu_k$ is true then

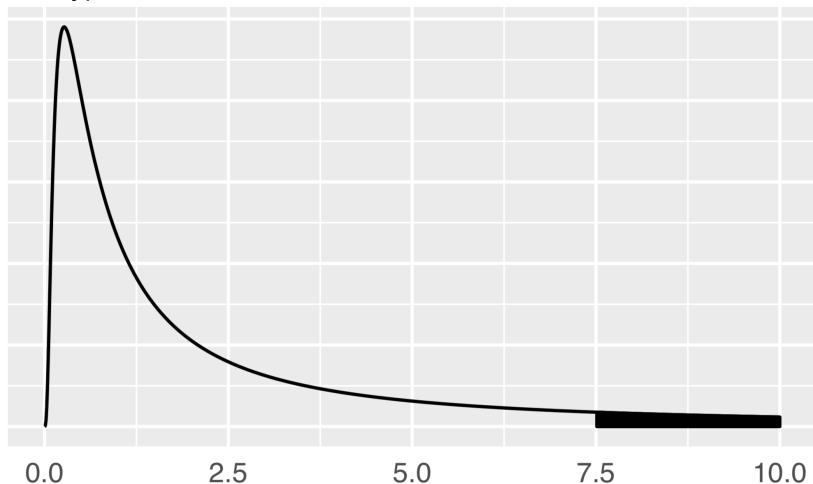
$$F = \frac{SSG/(k-1)}{SSE/(n-k)} = \frac{MSG}{MSE}$$

has an F distribution with $k-1$ and $n-k$ degrees of freedom.

We would reject the null hypothesis if $F \geq F_{1-\alpha, k-1, n-k}$, and as before we can find the probability of getting an F statistic this high or higher by using the RStudio code: **1-pf(F,k-1,n-k)**

F-distribution

The F -distribution is a probability density function over non-negative numbers. P-values are strictly calculated from the right tail – hence **large** F statistics indicate evidence **against** the null hypothesis.



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Example 1, from Larsen & Marx

A study examined the impact of smoking on heart rate by monitoring six each of nonsmokers, light smokers, moderate smokers, and heavy smokers under sustained physical exercise. Their heart rates were measured after resting for three minutes. Are the differences among the $\bar{Y}_j$ s significant? Let $\alpha = 0.05$.

	Nonsmokers	Light	Medium	Heavy
	69	55	66	91
	52	60	81	72
	71	78	70	81
	58	58	77	67
	59	62	57	95
	65	66	79	84
T_j	374	379	430	490
$\bar{Y}_{ij}$	62.3	63.2	71.7	81.7

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Adapted from notes by Dr. Gray & Dr. Roualdes.